

Violent crimes and homicide in New York City: The role of weather and pollution[☆]

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ABSTRACT

This paper studies the effect of weather and air pollution on violent crimes and homicide. We use all crimes committed in New York City between 2006 and 2020 and match each individual crime to the measures of the nearest weather, pollution station respectively. Our results show that neither weather nor pollutants have a contemporaneous effect of on homicide. In contrast, higher concentrations of carbon monoxide increase violent crimes in a U-shaped relation, that more rainfall linearly reduces violent crimes, and that sulphur dioxide has an inverted-U shaped relationship with violent crimes. Our results have implications for policing and environmental regulation.

1. Introduction

A large literature studies the relationship between weather – mainly temperature and rainfall – and crime^{1–3} and a growing literature studies the effect of air pollutants on crime.^{4,5}

A weakness of most studies is that they use low-frequency observations, i.e. monthly or annual data, which can lead to misleading results and that they do not include both drivers simultaneously, which can lead to classic omitted variable bias.⁶ This paper studies the relationship between weather variables and air pollutants with homicide and violent crimes more generally, using hourly crime observations over 15 years in New York City. We match each individual crime to the weather and pollution measures of the nearest weather, pollution station respectively. Compared to most papers in the literature,^{5,7,8} our data set allows for a finer spatial matching of individual crimes to stations and a higher frequency (hourly data) compared to, for example, county- or city-level annual data.^{7,8} Further, the underlying crime data allows for a detailed classification of crimes into, for example, homicide. We also use four types of pollutants (carbon monoxide, ozone, sulphur dioxide, nitrogen dioxide) rather than using an aggregate measure of pollutants (like particulate matter,^{8–10}; which allows to derive detailed policy recommendations.

Several findings stand out. We do not find a significant, contemporaneous effect of weather or pollutants on homicide in our sample. We

do find, however, that higher concentrations of carbon monoxide increase violent crimes in a U-shaped relation, that more rainfall linearly reduces violent crimes, and that sulphur dioxide has an inverted-U shaped relationship with violent crimes.

Our results are different to those in the related literature. Many papers find a positive relationship between temperature and violent crimes (e.g. Refs. 1,3,8–10. Our finding that temperature has a weakly significantly negative effect is more in line with the findings by Ranson.⁷ The negative effect of rainfall is in line with the related literature, while Herrnstadt et al.⁹ also find a positive, linear effect of carbon monoxide on violent crimes in Chicago. We are not aware of any other paper showing that sulphur dioxide has an effect on violent crimes.

This remainder of the paper is structured as follows. Section 2 explains the construction of our data set. Section 3 derives our hypothesis to be tested in section 4. Section 5 briefly concludes.

2. Data

We construct a data set by combining three individual data sets. First, we use crime data taken from the New York City OpenData platform in the historic NYPD Complaint data set. Second, we use weather data from NOAA's Local Climatological Data set (LCD). Third, we use air pollution data from the Environmental Protection Agency database.

We extract data on all violent crimes and two specifications of

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homicide, to be specified later. Crime data covers all reported felony, misdemeanour, and violation crimes to the NYPD between 2006 and 2020 (N = 2,496,887). Violent crimes are crimes in the following categories: (i) Murder and Non-negligent Manslaughter, (ii) Rape, (iii) Robbery, (iv) Felony Assault, (v) Burglary, (vi) Felony Sex Crimes, (vii) Assault 3 and Related Offenses, and (viii) Sex Crimes. Notice that this definition of violent crimes is different to the one used by the FBI in the Uniform Crime Reporting system. In line with the New York Penal Code, we additionally include burglary, sex crimes (§130.00 and §130.05), and assault 3 (§120.00 and §130.05). We then use two specifications of homicide. The first specification “Homicide” includes (i) Murder and Non-negligent Manslaughter, (ii) Homicide, negligent, unclassified, and (iii) Homicide, negligent, vehicle. The second specification “Murder” only includes Murder and Non-negligent Manslaughter and is, therefore, a narrower concept.

We use the following weather variables in our analysis: temperature measured in degrees Fahrenheit, rainfall measured in inches, visibility, which is the horizontal distance an object can be seen and identified (measured in miles), relative humidity measured in percent, wind speed measured in miles per hour, and wind direction measured in 45° degree bins. Further, we use four air pollutants. We measure the concentration of carbon monoxide in parts per million (ppm), sulphur dioxide in parts per billion (ppb), nitrogen dioxide in ppm, and ozone in ppm.¹

These pollutants are emitted by various sources. For carbon monoxide and nitrogen dioxide burning fuel by the about 2 million cars in NYC are the main cause. In contrast, industrial facilities and power plants emit sulphur dioxide. Some power plants in the NYC area, for example in Brooklyn and Manhattan, burn fuel oil, natural gas, and Kerosene. Finally, ozone is not emitted but is the outcome of a chemical process (combination of NOx, volatile organic compounds, temperature, and sunlight).

It is also important to notice that air pollution is not only an outdoor problem, but that indoor air pollution is typically even worse than outdoor air pollution (see Ref. 11).

For each crime in every category, we have the time (minute) and the geolocation (latitude, longitude) when the crime was committed. This allows to match the crime to the nearest weather station (either Central Park, LaGuardia Airport, or JFK Airport) and air pollution station (either Bronx, New York County (Manhattan), Queens, or Richmond). Because weather and air pollution data are measured at the hourly frequency, the match of each crime to weather and pollution data is rounded to the nearest hour. After this match has been established, we aggregate the three crime types to the hour frequency. Our variables used in the analysis are therefore the number of crimes committed in each hour from 2006 to 2020 (N = 122,705). Since some hours during this time have no crimes, we add zeros to the data set for hours without crimes.

Table 1 presents the descriptive statistics. In the data set, we have a total number of 686,023 violent crime observations, 3879 murders, and 3953 homicides. At the hourly frequency, we see that a maximum of eight murders and homicides were committed. The maximum of violent crimes per hour is 44. Average (dry bulb) temperature is 57° Fahrenheit and no rain falls on average. The relative humidity (a driver of wet bulb temperature) is 61%. Average wind speed was 9.37 miles per hour and visibility was about 9 miles on average. For the four pollutants, we find a large range and sufficient variation to exploit it in our statistical analysis. Average carbon monoxide concentration is 0.27 ppm (immediate danger level: 2,500 ppm), the average NO2 concentration is 18.55 ppm (200,000 ppb), for SO2 we find an average concentration of 0.69 ppb (100,000 ppb), and the average ozone concentration is 0.03 ppm (5 ppb).

¹ Notice that we do not include particulate matter (PMx) in our analysis, because PM includes all these pollutants likely leading to collinearity. Further, our analysis with individual pollutants will lead to more policy relevant conclusions about which pollutants to regulate.

Table 1
Summary statistics.

Variable	Observations	Mean	Std. Dev.	Min	Max
Violent Crime	2,496,887	0.27	0.75	0	44
Murder	2,496,887	0	0.04	0	8
Homicide	2,496,887	0	0.04	0	8
Other	2,496,887	0.53	1.20	0	83
<i>Weather</i>					
Temperature	927,515	57.20	17.38	−1	103
Rainfall	863,932	0	0.03	0	2.41
Rel. Humidity	927,452	61.39	19.33	6	100
Visibility	925,003	9.19	2.10	0	36.97
Wind Speed	900,029	9.37	5.96	0	750
<i>Pollution</i>					
CO	227,964	0.27	0.15	0	3.50
Ozone	444,887	0.03	0.02	0	0.13
NO2	338,150	18.55	12.19	0	153
SO2	204,006	0.69	0.98	0	22.10

To gain some intuition for the dynamics of violent crimes and homicide and the relationship with weather and pollution, Fig. 1 presents the within-day trend (Figure A.1 shows the same for murder) and Figs. 2 and 3 show the relationship with weather, pollution respectively (Figure A.2 and A.3 show the same for homicide).

Fig. 1 shows that violent crimes and homicides peak at about 6–8pm and that they have a low point around 6–8am in the morning. They steadily increase over the day. This relationship informs our estimation strategy discussed in the next section.

Further, Fig. 2 shows the relationship between violent crime and weather in the form of binned scatter plots. We find suggestive evidence for a non-linear relationship for temperature and visibility and a linear positive relationship with relative humidity and a negative relationship for rainfall. We find a peak for the temperature effect at around 70° Fahrenheit beyond which higher temperatures reduce violent crimes. Fig. 3 presents the same information for the pollution-violent crime relationship. We find a negative, linear relationship between violent crimes and carbon monoxide, SO2, and NO2. However, there is an inverted-U shaped relationship for ozone. The peak for the ozone effect is at a concentration of roughly 0.03 ppm.

Overall, this preliminary analysis suggests that there exists a relationship between violent crime, homicide, murder and weather as well pollution variables. For some variables, such as temperature and ozone, this relationship might be non-linear.

3. Hypotheses

The factors influencing violent crimes (incl. Murder and homicide) can broadly be classified into three categories: individual, sociocultural, and situational (e.g. Ref. 12. Individual factors include, for example, alcoholism, substance abuse, mental disorders (e.g. mood disorders), brain injuries, and poor parenting (e.g. Ref. 13. Sociocultural factors include factors such as norms and beliefs, accessibility of guns, policing, access to violent media entertainment, inequality, and poverty (e.g. Refs. 12,14. Situational factors, for example, are weather, noise, odors, peer pressure, pain and discomfort, and aggressive cues such as guns (e.g. Ref. 15. See also¹⁶ for an overview of the key theories explaining violent crime.

The idea that events and conditions affect criminal behaviour goes back a long time (see, e.g., Ref. 17. Theories have focused on how the environment directly and indirectly affects criminal behaviour.

¹⁸s General Affect Model suggests that the relationship between temperature and crime is positive, while¹⁹ and²⁰ in the Negative Affect Escape Model suggest an inverted-U shaped relationship. The idea behind these two models is that temperature affects aggression and arousal at

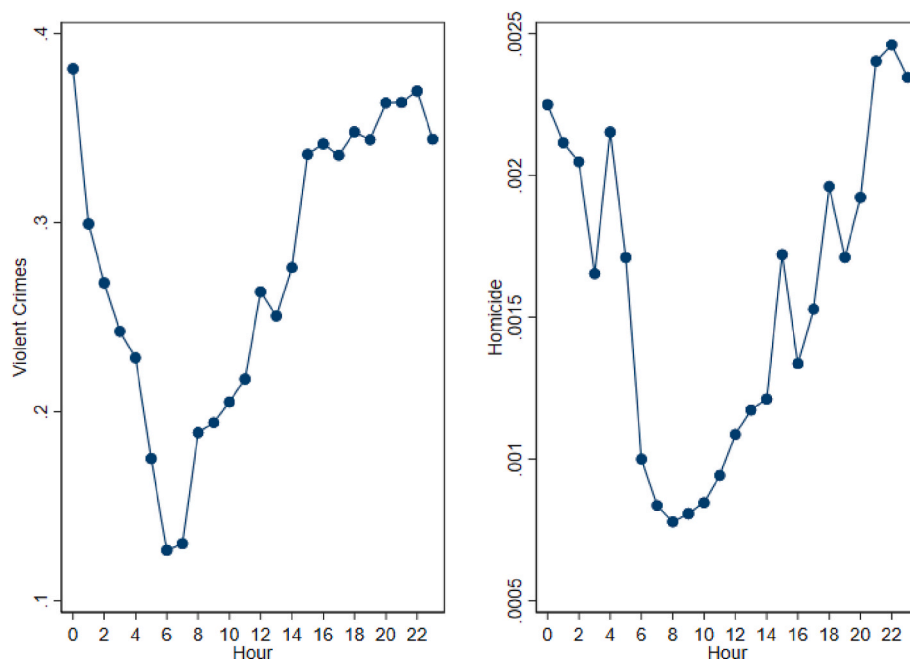


Fig. 1. Within-day trends of crimes.

Notes: Left-hand panel shows violent crimes and right-hand panel shows homicide.

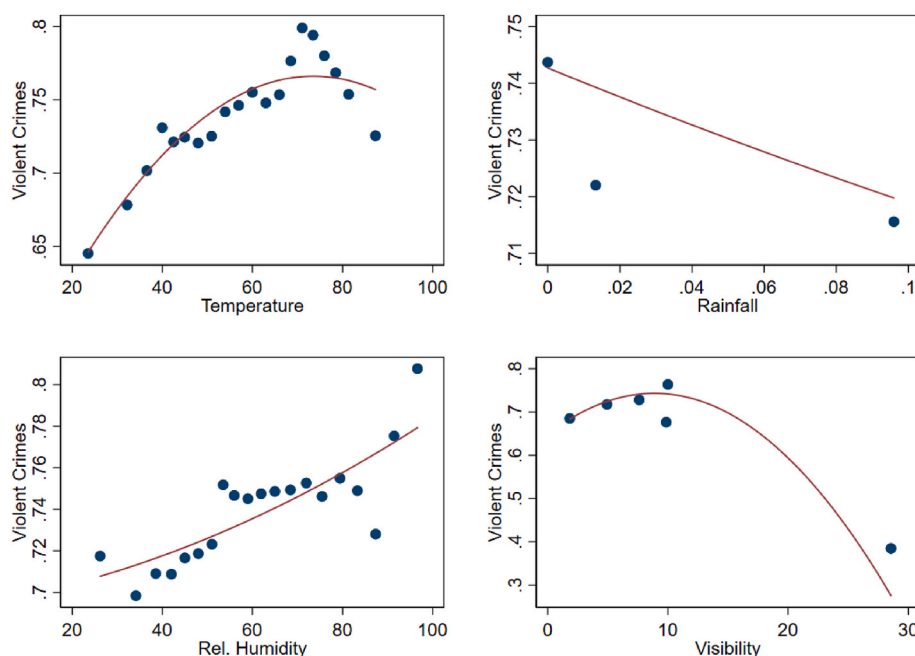


Fig. 2. Relationship between violent crime and weather.

Notes: Binned scatter plots for violent crime and temperature (top left), rainfall (top right), relative humidity (bottom left), and visibility (bottom right). Red line shows quadratic regression line.

the individual level, but that there is a point beyond which people prefer to avoid confrontation due to discomfort. Relatedly, pollutants will also affect behaviour at the individual level. A large literature documents the effect of carbon monoxide and ozone on brain activity.^{21,22} Pollutants small enough to penetrate the brain-blood barrier are able to enter the brain and lead to neuroinflammation, oxidative stress, and neurovascular unit dysfunction.²³ Additionally, pollutants also impair synaptic functioning²⁴ and cause neuron death.²⁵ These impacts on the brain then drive behaviour. For example, Lu et al.⁸ show that pollution affects anxiety and Rotton²⁶ shows that pollution affects tolerance

levels. For the other two pollutants, SO₂ and NO₂, we are not aware of effects on the brain, but they affect other health related outcomes, for example they affect the respiratory system (Kampa & Castanas, 2008). Importantly, these effects occur already after short-term exposure^{24,27} and also affect people indoors.¹¹

A different theory by Cohn & Felson²⁸, the *Routine Activity Theory*, postulates that a crime requires three things: a target, a motive, and the absence of a guardian. Within this theory, weather and pollution can affect, for example, the time we spend outdoors, which affects the target and guardian aspect of the theory.

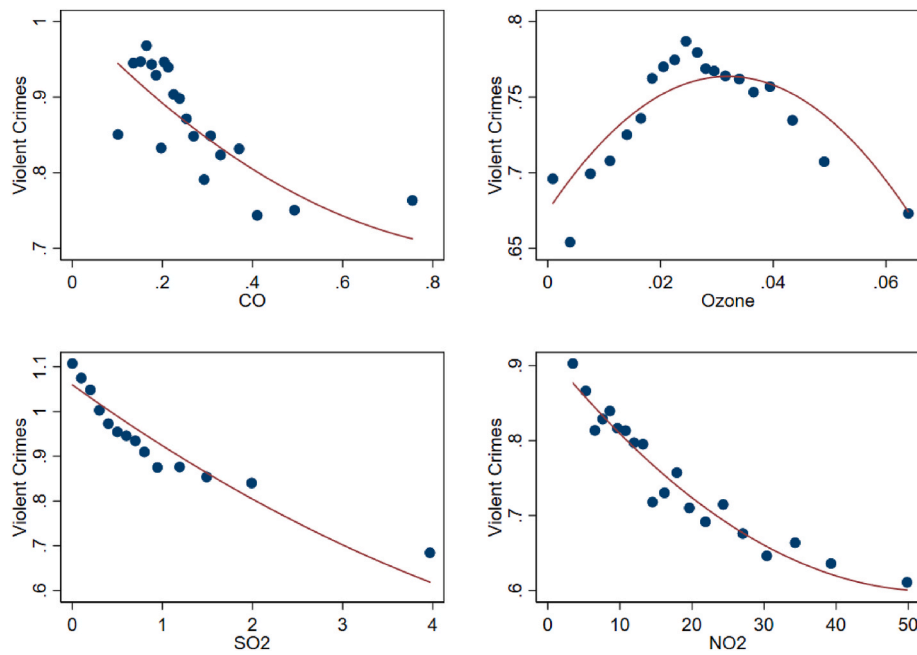


Fig. 3. Relationship between violent crime and pollution.

Notes: Binned scatter plots for violent crime and carbon monoxide (top left), ozone (top right), SO₂ (bottom left), and NO₂ (bottom right). Red line shows quadratic regression line.

In summation, we expect that higher temperatures lead to more crimes but the effect might be non-linear with a maximum. The effect of rainfall is ambiguous, because people might spend less time outdoors, but spending more time indoors could also increase violent crimes. We have no clear prior for variables like wind speed and direction as well as humidity. We would expect a positive effect of carbon monoxide and ozone on violent crimes, which again, could be non-linear. NO₂ and SO₂ could have an effect via the Routine Activity Theory (as they lead to respiratory effects) and lead to more or less crimes.

4. Empirical analysis

4.1. Methodology

Our estimation strategy relies on exploiting high-frequency fixed effects. Given the within-day trends including time fixed effects is important. Further, including fixed effects for weather and pollution station is important to capture constant spatial effects.

The key concern for our strategy is measurement error. In the absence of a valid instrumental variable (which affects pollution and weather but does not affect anything else), we use a different approach to address it. We use so-called weather and pollution “shocks”. The advantage is that these shocks do not have a systematic pattern and appear random. These shocks, intuitively, are the deviations of weather and pollution variables from their respective long-run trends. These are sometimes also referred to as “anomalies”. Technically, we construct weather and pollution shocks by first computing the year-month-day-hour trend of each variable and then calculating the deviation from this trend at the hourly frequency. These obtained shocks then should be not affected by measurement error (see Figures A.4 and A.5 in the appendix for the time series).

Having aggregated the data to the hourly frequency, we have generated a count data variable. Hence, we use a negative binomial regression due to significant overdispersion in the data.

We then estimate the following model:

$$Crime_{ijt} = \alpha + \beta f(Weather_{ijt}) + \delta g(Pollution_{ijt}) + \theta_i + \theta_j + \theta_t + \varepsilon_{ijt}, \quad (1)$$

where $Crime_{ijt}$ is the number of crimes at hour t in the weather-pollution station region i and j . We are interested in estimating the effect of our weather variables, β , and pollutants, δ . We allow the functions f and g to be non-linear and include squared terms to capture the potential non-linearity shown in the scatter plots. We include fixed effects for weather stations, θ_i , pollution stations, θ_j , as well as year, month, day, and hour fixed effects, which we all denoted by θ_t . All regressions use clustered standard errors at the hourly level.

Finally, delayed reporting of crimes could be a threat to our analysis. However, we find that only 3.2% of homicides and 2.7% of murders are reported with a delay (with an average delay of about 14 days). For violent crimes, we find that 23.9% are reported with a delay (with an average delay of about 15 days). Therefore, delays should not affect our results for our key variables of interest: murder and homicide, but our results for violent crimes should be interpreted with these reporting delays in mind.

4.2. Main results

Our estimation results are presented in Table 2. This table presents the results from estimating equation (1). Precisely, the table show the average marginal effects (or predictive margins) from a negative binomial regression. Intuitively, these numbers can be interpreted as follows. Take the coefficient of temperature (-0.0066) as an example: if temperatures increase by 1 (degree Celsius), violent crimes will fall by -0.0066 .

Our first key finding is that there is a U-shaped relationship of violent crime and carbon monoxide. The minimum is found at a CO concentration of 0.123 ppm, which is below the average value (0.27). This implies that for most of the support of the carbon monoxide variable, there exists a positive relationship and higher values of CO will lead to more violent crimes being committed. Further, there is an inverted-U shaped relationship with SO₂. We find a peak of 0.59 below its average of 0.69. Hence, most of the time the effect of higher SO₂ concentrations will reduce the number of violent crimes.

We also find that higher temperatures have a weakly significant ($p < 0.1$) negative effect on violent crimes. Further, we find that rainfall has a negative effect ($p < 0.05$) on violent crimes.

Table 2
Regression results.

Variable	Violent Crime	Homicide	Murder
Temperature	−0.0066* (0.0035)	0.0000 (0.0002)	0.0000 (0.0002)
Rainfall	−0.3629** (0.1830)	0.0068 (0.0123)	0.0068 (0.0127)
Visibility	0.0034 (0.0047)	−0.0001 (0.0002)	−0.0001 (0.0002)
Wind Speed	−0.0006 (0.0014)	0.0001 (0.0001)	0.0001 (0.0001)
Rel. Humidity	0.0003 (0.0012)	0.0000 (0.0001)	0.0000 (0.0001)
CO	−0.0882* (0.0475)	0.0045 (0.0030)	0.0044 (0.0030)
Ozone	−0.3803 (0.7392)	−0.0591 (0.0584)	−0.0500 (0.0614)
NO2	−0.0009 (0.0015)	0.0000 (0.0001)	0.0000 (0.0001)
SO2	0.0105 (0.0124)	−0.0005 (0.0005)	−0.0004 (0.0005)
Temperature ²	0.0001 (0.0003)	0.0000 (0.0000)	0.0000 (0.0000)
Rainfall ²	−0.0031 (0.3561)	−0.0050 (0.0077)	−0.0046 (0.0078)
Visibility ²	0.0000 (0.0008)	0.0000 (0.0000)	0.0000 (0.0000)
Wind Speed ²	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Rel. Humidity ²	0.0001 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
CO ²	0.3587* (0.1873)	0.0054 (0.0049)	0.0054 (0.0045)
Ozone ²	1.6031 (76.8051)	3.7227 (5.0358)	3.6260 (4.9729)
NO2 ²	−0.0001 (0.0002)	0.0000* (0.0000)	0.0000 (0.0000)
SO2 ²	−0.0089** (0.0037)	0.0001 (0.0002)	0.0001 (0.0002)
Observations	122,705	122,705	122,705
Pseudo R ²	0.06	0.04	0.04

Notes: Table shows average marginal effects from a negative binomial regression. Dependent variable: number of crimes per hour. Regression uses negative binomial model with pollution and weather station fixed effects as well as fixed effects for year, month, day, and hour. Standard errors are clustered at the hourly level. Constant not shown. Significance levels: *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$.

Our second key finding is that neither weather nor pollution affect homicide or murder. This implies that there is statistically significant *contemporaneous* effect of weather or pollution variables on homicide or murder. However, this result still leaves the possibility that there is a cumulative effect of these variables. The problem to answer this question is that we do not know which homicide or murder was premeditated and which was not. Therefore, we cannot investigate this question with our data.

Appendix

The results for violent crimes are in line with our prior hypotheses. The effect of carbon monoxide on violent crimes is explained by the effects of these pollutants on brain and, consequently, behaviour. Since it is an odourless and invisible gas, it should not affect people's behaviour in line with the Routine Activity Theory. However, the effect of rainfall and SO2 are in line with this theory. Our results for rainfall suggest that people might spend more time indoors and are less likely to become the victim of a crime (less targets available). SO2 affects, for example, the respiratory system which – we postulate – should have a similar effect compared to rainfall in that people stay home, if the concentration of SO2 becomes too high (above the threshold of 0.59 ppm).

Further, our results contrast the related literature. For example, the literature typically finds a positive effect of temperature on violent crimes (e.g. Refs. 1,3,8–10). Our finding is more in line with the result by Ranson⁷, showing a negative effect of higher temperatures on murder, manslaughter, and other violent crime types in a sample of US counties between 1980 and 2009 using annual data. In contrast, our finding of a negative effect of rainfall on violent crimes is similar to the literature (e.g. Refs. 1,9,10). Herrnstadt et al.⁹ find that carbon monoxide has a positive, linear effect on violent crimes in Chicago. The difference in our paper is that we allow for a non-linear effect. Finally, we are not aware of a paper showing that SO2 has an effect on violent crime.

5. Conclusion

This paper studies the effect of weather and air pollution on homicide and violent crimes. Using a rich data set of crimes in New York City over 15 years combined with hourly weather and air pollution measures from multiple stations within New York City, we study this relationship using a negative binomial regression.

Our results are in line with existing theories of crime and environmental factors.^{18,20,28} We find that rainfall and SO2 have a negative effect on violent crimes. In contrast, higher concentrations of carbon monoxide increase violent crimes. We do not find any significant contemporaneous effect of weather or air pollutants on homicide.

The results in the paper carry implications for policing and environmental regulation. Reducing the concentration of carbon monoxide appears to be an important policy to reduce violent crimes. Further, the findings can also inform the allocation of police resources as well as increase the accuracy of forecasting algorithms.^{29,30}

Data availability

Upon request from author.

Declaration of competing interest

None and no funding received.

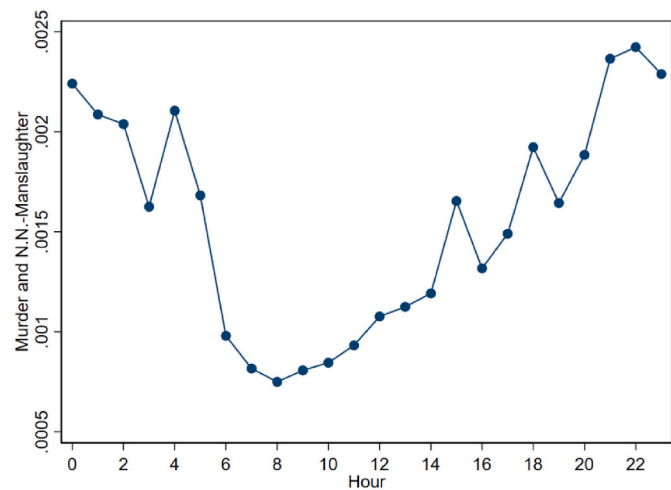


Fig. A.1. Within-Day Trends of Murder. .

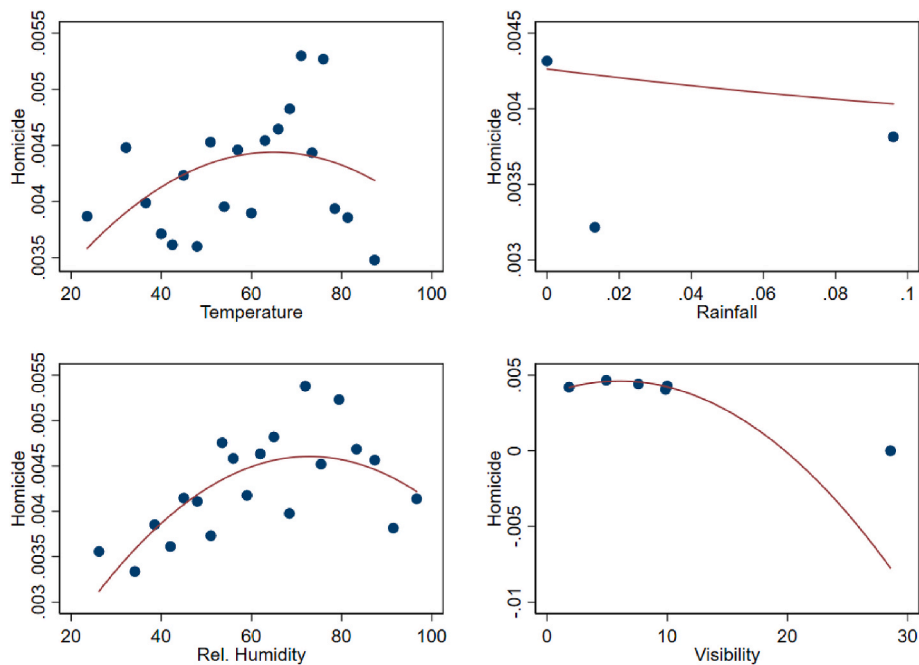


Fig. A.2. Relationship between Homicide and Weather. .

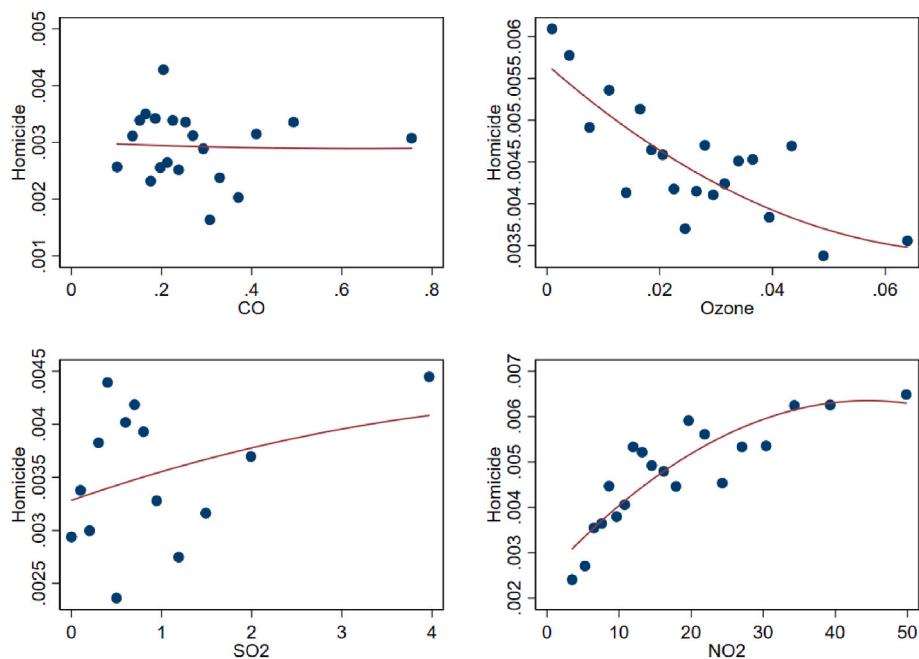


Fig. A.3. Relationship between Homicide and Pollution. .

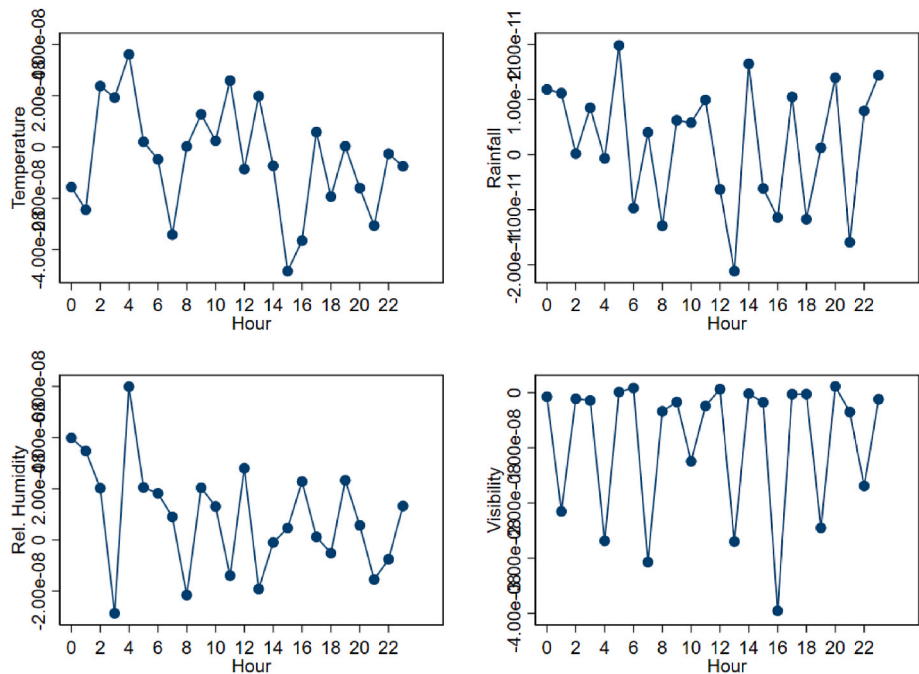


Fig. A.4. Time Series of Weather Shocks. .

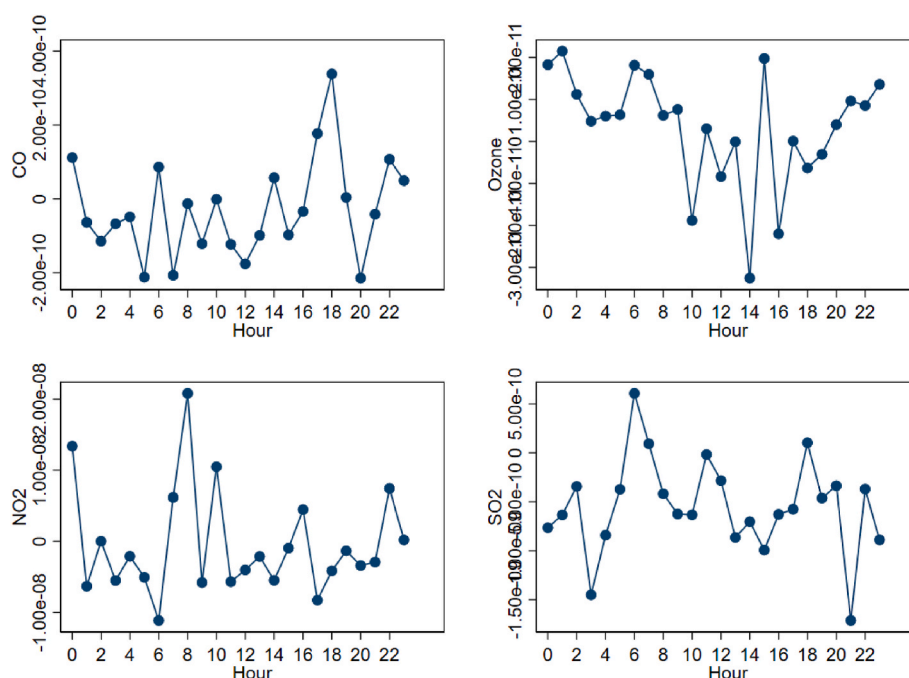


Fig. A.5. Time Series of Pollution Shocks. .

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